Matrix Decompositions: LU, Cholesky, Full Rank, and SVD

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§1.1 Why decompositions are algorithms

A decomposition rewrites a matrix into factors that are easier to compute with. LU reduces a system to triangular solves; Cholesky uses positive definiteness; full-rank decomposition exposes rank; SVD exposes singular directions.

§1.2 Doolittle LU

If $A \in \mathbb{C}^{n\times n}$ can be written as

$$A = LU,$$

where L is unit lower triangular and U is upper triangular, this is the Doolittle or LU decomposition.

§1.3 Existence and uniqueness

If the first $n-1$ leading principal minors of A are nonzero, then $A = LU$ exists. If A is nonsingular, the decomposition is unique. The condition says that Gaussian elimination without row exchange never meets a zero pivot.

§1.4 Elimination matrices

At step k, an elimination matrix L_k zeros entries below the pivot. After all steps,

$$L_{n-1}\cdots L_1 A = U, \qquad A = L_1^{-1}\cdots L_{n-1}^{-1}U.$$

The product of the inverses is the lower triangular factor.

§1.5 Doolittle formulas

Comparing $A = LU$ gives

$$u_{kj} = a_{kj} - \sum_{t=1}^{k-1} l_{kt}u_{tj}, \qquad j \geq k,$$

$$l_{ik} = \frac{1}{u_{kk}}\left(a_{ik} - \sum_{t=1}^{k-1} l_{it}u_{tk}\right), \qquad i > k.$$

The pivot u_{kk} must be nonzero.

§1.6 Pivoting

If a pivot is zero or small, row exchange is needed. With partial pivoting,

$$PA = LU,$$

where P is a permutation matrix. Pivoting controls large multipliers and improves numerical stability.

§1.7 Storage of LU

The upper triangle stores U; the strictly lower triangle stores the multipliers l_{ij}. The diagonal of L is omitted because it is all ones.

§1.8 Cholesky decomposition

If A is Hermitian positive definite, there is a unique lower triangular L with positive diagonal such that

$$A = LL^H.$$

Solving $Ax = b$ becomes

$$Ly = b, \qquad L^H x = y.$$

Cholesky costs about $\frac{1}{3}n^3 + O(n^2)$, roughly half of LU.

§1.9 Cholesky example

For

$$A = \begin{bmatrix} 4 & 2 \\ 2 & 5 \end{bmatrix},$$

we obtain

$$L = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}, \qquad A = LL^T.$$

§1.10 Full-rank decomposition

If $\operatorname{rank} A = r > 0$, a full-rank decomposition is

$$A = FG, \qquad F \in \mathbb{C}^{m\times r}, \quad G \in \mathbb{C}^{r\times n},$$

with both factors rank r. It always exists and is not unique.

§1.11 Hermite standard form

Row operations reduce A to a Hermite-style form H. If pivot columns are $j_1, \dots, j_r$, choose F as the corresponding columns of A, and G as the nonzero rows of H. Then $A = FG$.

§1.12 Singular values

The singular values of A are

$$\sigma_i = \sqrt{\lambda_i(A^H A)}.$$

Unitary equivalence preserves singular values.

§1.13 Singular value decomposition

If rank $A = r$, then

$$A = U \begin{bmatrix} \Sigma & 0 \\ 0 & 0 \end{bmatrix} V^H, \qquad \Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_r).$$

SVD says that A rotates, stretches along orthogonal axes, and rotates again.

§1.14 SVD geometry

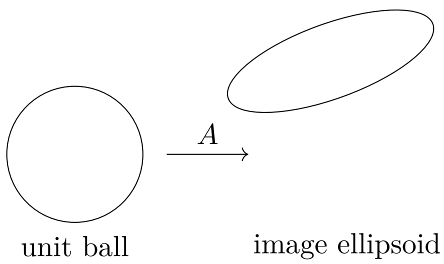

Figure 1.1: SVD describes the image of the unit ball as an ellipsoid.

§1.15 Checklist

LU	general square systems
$PA = LU$	pivoted stability
$A = LL^H$	positive definite systems
$A = FG$	rank structure
$A = U\Sigma V^H$	singular directions